COURSE CODE: DECAP538
COURSE NAME: ALGORITHM DESIGN AND ANALYSIS

Short Questions(2 Marks):

Q.1 What are elementary data structures?
Q.2 Define computational model.
Q.3 What is algorithm analysis?
Q.4 Differentiate between best case and worst case behaviour.
Q.5 What is Big O notation?
Q.6 Define recursion.
Q.7 What are recurrence relations?
Q.8 Explain divide and conquer method.
Q.9 What is binary search?
Q.10 Define merge sort.
Q.11 What is quick sort?
Q.12 Explain arithmetic with large integers.
Q.13 What is greedy method?
Q.14 Define knapsack problem.
Q.15 What is a minimum spanning tree?
Q.16 State Prim's algorithm.
Q.17 State Kruskal's algorithm.
Q.18 What is single source shortest path problem?
Q.19 Define dynamic programming.
Q.20 What is chained matrix multiplication?
Q.21 Explain optimal storage on tapes.
Q.22 What is all-pairs shortest path problem?
Q.23 Define optimal binary search tree.
Q.24 What is backtracking?
Q.25 Explain the 8-queens problem.
Q.26 What is graph coloring?
Q.27 Define Hamiltonian cycle.
Q.28 What is branch and bound method?
Q.29 Explain 0/1 knapsack problem.
Q.30 What is travelling salesperson problem?
Q.31 Define pattern matching.
Q.32 What is brute force pattern matching?
Q.33 Explain KMP algorithm.
Q.34 What is Boyer Moore algorithm?
Q.35 Define Huffman coding.
Q.36 What is data compression?
Q.37 Explain lower bound theory.
Q.38 What is a comparison tree?
Q.39 Define oracle and adversary arguments.

Q.40 What are reductions in lower bound theory?
Q.41 Define approximation algorithms.
Q.42 What is task scheduling?
Q.43 Explain bin packing problem.
Q.44 What are intractable problems?
Q.45 Define NP-complete problems.
Q.46 What are NP-hard problems?
Q.47 Explain Cook's theorem.
Q.48 What is problem reduction?

Long Questions (10 Marks):

Q.1 Explain elementary data structures and basic computational models in detail.
Q.2 Discuss the analysis of algorithms with best case, average case, and worst-case behaviour.
Q.3 Explain asymptotic notations and Big O notation with examples.
Q.4 Discuss recursion and recurrence relations used in recursive algorithms.
Q.5 Explain divide and conquer strategy with suitable examples.
Q.6 Describe binary search algorithm with its complexity analysis.
Q.7 Explain merge sort and quick sort algorithms in detail.
Q.8 Discuss arithmetic operations with large integers.
Q.9 Explain greedy method and its applications.
Q.10 Solve the knapsack problem using greedy method.
Q.11 Discuss Prim's and Kruskal's algorithms for minimum spanning trees.
Q.12 Explain single source shortest path algorithms.
Q.13 Discuss dynamic programming and its general method.
Q.14 Explain chained matrix multiplication using dynamic programming.
Q.15 Discuss optimal storage on tapes problem.
Q.16 Explain all-pairs shortest paths problem.
Q.17 Discuss the construction of optimal binary search trees.
Q.18 Explain backtracking technique with suitable examples.
Q.19 Discuss the 8-queens problem using backtracking.
Q.20 Explain graph coloring and Hamiltonian cycles.
Q.21 Discuss branch and bound method in detail.
Q.22 Solve the 0/1 knapsack problem using branch and bound.
Q.23 Explain travelling salesperson problem.
Q.24 Discuss brute force, KMP, and Boyer Moore pattern matching algorithms.
Q.25 Explain Huffman coding and data compression techniques.
Q.26 Discuss lower bound theory and comparison tree method.
Q.27 Explain oracle and adversary arguments with examples.
Q.28 Discuss lower bounds through reductions.
Q.29 Explain approximation algorithms with task scheduling and bin packing problems.
Q.30 Discuss intractable problems, NP-hard and NP-complete problems in detail.
Q.31 Explain Cook's theorem and problem reduction techniques.